

Non-contextual meanings and functional readings of *wh*-questions

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1 Introduction

Three readings of a *wh*-question with a quantifier:

- (1) Q: Who does everyone love?
A: Sue (individual reading)
A': Their mother. (functional reading)
A'': John loves Sue; Bill loves Mary; Tim loves Amy. (list reading)

Proposal by many linguists like Groenendijk and Stokhof (1984), Engdahl (1980, 1986), and Chierchia (1993), Hornstein (1995), Sharvit (1997,1999), Dayal (1996, 2016), Xiang (2023), etc.:

The trace of a *wh*-phrase in a functional reading question carries two or more indexes, one of which is bound by a *wh*-phrase and the other(s) by (a) quantifier(s).

- (2) Who_{*i*} does everyone_{*j*} love t_{*i*}^{*j*}?
[who *f* does [everyone_{*x*} love *f*(*x*)]]

Problems

- In natural language syntax, it has been assumed that a movement of a phrase leaves one trace with one index.
- If the *wh*-phrase denotes a function, it should be *what*, not *who*.
- If there is more than one quantifier in a sentence, the trace of a *wh*-phrase may need three or more indexes.

- (3) a. Q: To whom does every writer dedicate a book?
A: To the person that inspired her/him to write it.
- b. $[[\text{to whom}]_i [\text{every writer}]_j [\text{a book}]_k [t_j \text{ dedicates } t_k t_i^{j,k}]]$
- c. $\{p \mid \exists f[p = \forall x[\text{writer}(x) \rightarrow \exists y[\text{book}(y) \wedge \text{dedicate}(x, y, f(x, y))]]]\}$

-
- In the short answer *Their mother* in (1), the pronoun is free.

(4) $\llbracket \text{Their mother} \rrbracket = \iota x[mother(x, y)]$
Cf. Required interpretation: $\lambda y \iota x[mother(x, y)]$

Purpose of this paper

To propose an analysis that assumes a single-indexed trace and that lacks the problems

2 Necessity of non-contextual meanings

2.1 Problems with binding relations

Reconstruction required:

In *wh*-questions, if a preposed phrase includes a DP that is subject to Binding Conditions (BCs), it must be reconstructed into the positions of its trace.

- (5) a. *Which of *each other*'s friends did she remind t that *they* saw Bill?
b. Which of *each other*'s friends did *they* remind t that she saw Bill?
c. Which of *each other*'s friends did she remind Bill that *they* saw t ?
- (6) *Whose examination of *John* did *he* fear?
⇒ *did *he* fear whose examination of *John* ?

It is reasonable to assume that **all preposed *wh*-phrases must be reconstructed.**

Intervention effects:

A *wh*-phrase cannot occur after a focused phrase.

(7) a. ??When did [only John_F] visit which museum?

b. Which museum did [only John_F] visit?

(8) Which of *each other*'s friends did she remind [only Bill_F] that *they* saw t?

⇒ We must get the reconstruction effect without syntactic reconstruction.

2.2 Non-contextual meanings

Intervention effects + Reconstruction effects $\Rightarrow$

- A preposed *wh*-phrase must be interpreted so that it can get the scope of its trace.

Question 1: How is a preposed *wh*-phrase interpreted?

- If a preposed *wh*-phrase has the scope of its trace, it is likely to behave like a *wh*-phrase in situ.

Question 2: How is a *wh*-phrase in situ associated with the Mood?

Answer to Question 1: Non-contextual meanings

- An expression is interpreted w.r.t. a linguistic context g^*w^* , where g^* and w^* are an assignment and a possible world given in the context.

$\llbracket \dots \rrbracket^{g^*w^*}$

- The meaning of an expression with the feature $[+wh]$ is a non-contextual meaning.

non-contextual meaning of an expression: $\llbracket \dots^{[+wh]} \rrbracket^{g^*w^*} = \lambda \langle g, w \rangle \llbracket \dots \rrbracket^{gw}$

- A *wh*-phrase and its trace carry the $[+wh]$ feature.
- A non-contextual meaning is contextualized with respect to a local context if the expression is in an IP, which denotes a proposition.

$$\llbracket \hat{[IP...]} \rrbracket^{g^*w^*} = \lambda w' \llbracket [...] \rrbracket^{g^*w'}$$

$$\llbracket [DP...]^{[+wh]} \rrbracket^{g^*w'} = \lambda \langle g, w \rangle \llbracket [DP...] \rrbracket^{gw}(\langle g^*, w' \rangle) = \llbracket [DP...] \rrbracket^{g^*w'}$$

(9)	$[_{MP}$	<i>Wh-phrase</i> _{<i>i</i>}	...	$[_{IP}$	...	$t_i^{[+wh]}$	...	<i>wh-phrase</i>	...]
		non-contextual				non-contextual		non-contextual	
		not contextualized				contextualized		contextualized	

- For the trace of a *wh*-phrase, we need a new assignment that assigns a non-contextual meaning.

Contextual assignments: $f, g, \dots$

Non-contextual assignments: $\vec{f}, \vec{g}, \dots$

$$(10) \quad \llbracket t_k^{[+wh]} \rrbracket^{\vec{g}g^*w'} = \vec{g}(k)(\langle g^*, w' \rangle) \text{ (or preferably } \vec{g}(k)^{g^*w'} = g^*(k)$$

- (5) c. $[_{MP}[_{DP}\text{Which of } \textit{each other}_j\text{'s friends}]_k^{[+wh]} \text{ did } \wedge [\text{she}_i \text{ remind Bill } \wedge [\text{that } \textit{they}_j \text{ saw } t_k^{[+wh]}]]]]$
- i) $\lambda\langle g, w \rangle \llbracket \text{Which of each other's friends} \rrbracket^{\vec{g}gw} (= \mathbf{A})$
- ii) $\llbracket \text{did } \wedge [\text{she}_i \text{ remind Bill } \wedge [\text{that } \textit{they}_j \text{ saw } t_k^{[+wh]}]]] \rrbracket^{\vec{g}g^*w^*} =$
 $\llbracket \text{did} \rrbracket^{g^*w^*} (\lambda w' [\llbracket \text{remind} \rrbracket^{g^*w'} (g^*(i), \textit{bill}, \lambda w'' [\llbracket \text{saw} \rrbracket^{g^*w''} (g^*(j), \vec{g}(k)(\langle g^*, w'' \rangle)))])])$
- i) + ii) $\llbracket [\text{Which of } \textit{each other}\text{'s friends}]_k \text{ did } \wedge [\text{she remind Bill } \wedge [\text{that } \textit{they} \text{ saw } t_k]]] \rrbracket^{\vec{g}g^*w^*}$
 $= \lambda \vec{x} \llbracket \llbracket \text{did } \wedge [\text{she}_i \text{ remind Bill } \wedge [\text{that } \textit{they}_j \text{ saw } t_k^{[+wh]}]]] \rrbracket^{\vec{g}[k/\vec{x}]g^*w^*} \rrbracket (\mathbf{A})$
 $= \llbracket \llbracket \text{did} \rrbracket^{g^*w^*} (\lambda w' [\llbracket \text{remind} \rrbracket^{g^*w'} (g^*(i), \textit{bill}, \lambda w'' [\llbracket \text{saw} \rrbracket^{g^*w''} (g^*(j), \mathbf{A}(\langle g^*, w'' \rangle)))])]) \rrbracket$
 $= \llbracket \llbracket \text{did} \rrbracket^{g^*w^*} (\lambda w' [\llbracket \text{remind} \rrbracket^{g^*w'} (g^*(i), \textit{bill},$
 $\lambda w'' [\llbracket \text{saw} \rrbracket^{g^*w''} (g^*(j), \llbracket \text{Which of each other's friends} \rrbracket^{g^*, w''}))])]) \rrbracket$
 $= \llbracket \text{did } \wedge [\text{she remind Bill } \wedge [\text{that } \textit{they} \text{ saw } \textit{Which of each other's friends}]] \rrbracket^{\vec{g}g^*w^*}$

- We get the effect of reconstruction with the help of non-contextual meanings.
- The anaphor *each other* is semantically bound.

(The notion of semantic binding needs to be defined. See Yeom's (2025) talk "Semantic binding over syntactic binding".)

- The result: All *wh*-phrases are *wh*-phrase in situ.

Semantics of the interrogative Mood

- The meaning of *did*: $\llbracket \text{did} \rrbracket : \text{a proposition} \rightarrow \text{a pair of propositions}$
 $\llbracket \text{did} \rrbracket(\llbracket \hat{\phi} \rrbracket) = \{ \langle p, \neg p \rangle \mid p \in \llbracket \hat{\phi} \rrbracket \text{ or } p = \llbracket \hat{\phi} \rrbracket \}$

- (11) a. Did John laugh?
 $\{ \langle \lambda w \llbracket \text{John laughed} \rrbracket^w, \lambda w [\neg [\llbracket \text{John laughed} \rrbracket^w]] \rangle \}$
- b. Who did John love? $\Rightarrow$ did John love Who
 $\{ \langle \lambda w [\llbracket \text{loved} \rrbracket^w(d)(john)], \lambda w [\neg [\llbracket \text{loved} \rrbracket^w(d)(john)]] \rangle \mid d \in D \}$
 $= \textcolor{blue}{POW}(\{ \lambda w [\llbracket \text{loved} \rrbracket^w(d)(john)] \mid d \in D \})$
- e.g. $\{ \langle A, \neg A \rangle, \langle B, \neg B \rangle \} = POW(\{A, B\})$
(the number of possible answers = 4)

Question 2: How is a *wh*-phrase in situ associated with the Mood?

Answer to Question 2: Alternative non-contextual meanings

- A *wh*-word *which* generates a set of alternatives.
- Some *wh*-words can be assumed to include some descriptive content. I ignore the descriptive content for the time being.

(12) who = which person;
where = which place;
when = which time
(*which* here is number-neutral)

- The alternatives also denote non-contextual meanings.

- (13) $\text{Who}_j^{[+wh]}$ does John believe every boy_i loves $t_j^{[+wh]}$?
 $\Rightarrow \llbracket [{}_M \mathbf{does}] \wedge [\text{John believe} \wedge [\text{every boy}]_i \text{ loves } \text{Who}_j^{[+wh]}] \rrbracket \vec{g}g^*w^*$
- a. $??POW(\{\lambda w'[\text{John believe}^{w'}(\lambda w''[\forall x[\text{boy}^{w''}(x) \rightarrow \text{love}^{w''}(x, z)]]]) \mid z \in \text{Alt}(\llbracket \text{who}_j \rrbracket \vec{g}g^*[j/y]w^*)\})$
 main context: $\llbracket \text{who}_j \rrbracket \vec{g}g^*[j/y]w^* = g^*(j)$
- b. $??POW(\{\lambda w'[\text{John believe}^{w'}(\lambda w''[\forall x[\text{boy}^{w''}(x) \rightarrow \text{love}^{w''}(x, z)]]]) \mid z \in \text{Alt}(\llbracket \text{who}_j \rrbracket \vec{g}g^*[i/x][j/y]w'')\})$
 local context: $\llbracket \text{who}_j \rrbracket \vec{g}g^*[i/x][j/y]w^* = g^*[i/x][j/y](j) = y$
- c. $\checkmark POW(\{\lambda w'[\text{John believe}^{w'}(\lambda w''[\forall x[\text{boy}^{w''}(x) \rightarrow \text{love}^{w''}(x, \vec{z}^{g^*[i/x][j/y]w''})]]]) \mid \vec{z} \in \text{Alt}(\llbracket \text{who}_j^{[+wh]} \rrbracket \vec{g}[j/\vec{y}]g^*w^*)\})$
 non-contextual: $\llbracket \text{who}_j^{[+wh]} \rrbracket \vec{g}[j/\vec{y}]g^*w^* = \vec{g}(j)$

- (13a) leads to the individual reading.
- (13b) is not available because i) x in $g^*[i/x][j/y]$ is free and ii) something introduced in the scope of a universal quantifier or a modal operator is not available in the main context, but the interrogative Mood is in the matrix clause.

- (14) a. ??Every man has *a car*. *It* is not fast.
b. ??Amy thinks Bob has *a car*. *It* is not fast.

- (13c) is the only possible reading.

Results:

i) A preposed *wh*-phrase gets the effect of reconstruction and is associated with the Mood by denoting a set of alternative non-contextual meanings.

Question 3: Why is a *wh*-phrase preposed?

ii) A non-contextual meaning is scopeless.

Question 4: How is the scope of a *wh*-phrase determined?

2.3 Why is a *wh*-phrase preposed?

- In languages like Korean and Japanese, no *wh*-movement is necessary.
- In English, if there is more than one *wh*-phrase in a question, only one is preposed.
- The scope of a *wh*-phrase in situ is not subject to syntactic islands, though a *wh*-movement is.

(15) Which linguist will be offended if [we invite *which philosopher*]?

- A preposed *wh*-phrase does not have the scope that a DP normally gets in the surface position. A *wh*-movement is not for the scope of the *wh*-phrase.

(16) A: John is insane. He believes {a hero in a movie, an imaginary creature} attacked him.

B: {Who, What} does John believe attacked him?

A: {Iron Man, A unicorn}.

- The classical analyses of *wh*-questions like Hamblin (1973) and Karttunen (1977) are not quite right.

(17) $\{p \mid \exists x[\text{person/animal}(x) \wedge p = \hat{\text{believe}}(\text{john}, \hat{\text{attack}}(x, \text{john}))]\}$

Answer to Question 3: In English, a *wh*-movement is to indicate the scope of a question, not the scope of the *wh*-phrase.

2.4 How is the scope of a *wh*-phrase determined?

- A *wh*-phrase is generally taken to be an indefinite.
- An indefinite has three interpretations.

- (18) If *a relative of mine* dies, I will inherit a house.
- a. If there is *a relative of mine* and he or she dies, I will inherit a house.
 - b. (I live with some relatives of mine.) If *one of them* dies,
 - c. (I live with a relative of mine.) If *she* dies,

(19) $\llbracket [a \ \alpha] \ \beta \rrbracket =$

a. $\exists x[\llbracket \alpha \rrbracket(x) \wedge \llbracket \beta \rrbracket(x)]$ (quantificational use)

b. $\ll \exists X[X \neq \emptyset \wedge \forall x \in X[\llbracket \alpha \rrbracket(x)]] \gg \exists x' \in X[\llbracket \beta \rrbracket(x')]$ (split scope)
 $\ll \dots \gg$: presupposition

Only the **descriptive content** is presupposed.

c. $\ll \exists x[\llbracket \alpha \rrbracket(x)] \gg \llbracket \beta \rrbracket(x)$ (specific use)

Both the **quantificational force** and the **descriptive content** are presupposed.

(See Yeom's Ph.D. Thesis (1997/1998) & Split Scope paper (2021).)

- A *wh*-phrase is a specific indefinite, whose scope is determined by presupposition projection of the quantificational force and the descriptive content.

(20) a. If John stops smoking, he will get healthier.

= John has been smoking. If John stops it, he will get healthier.

b. If *a relative of mine* dies, I will inherit a house.

= There is a relative of mine. If she dies, I will inherit a house.

- A *wh*-phrase as a specific indefinite tends to have wide scope, but despite the tendency, it can have narrow or intermediate scope if it is required.

- (21) Every boy drove *his* car.
 = Every boy had a car and drove it.
 ≠ Everyone had a car. Every boy drove it.

Interpretation of a *wh*-phrase as a specific indefinite:

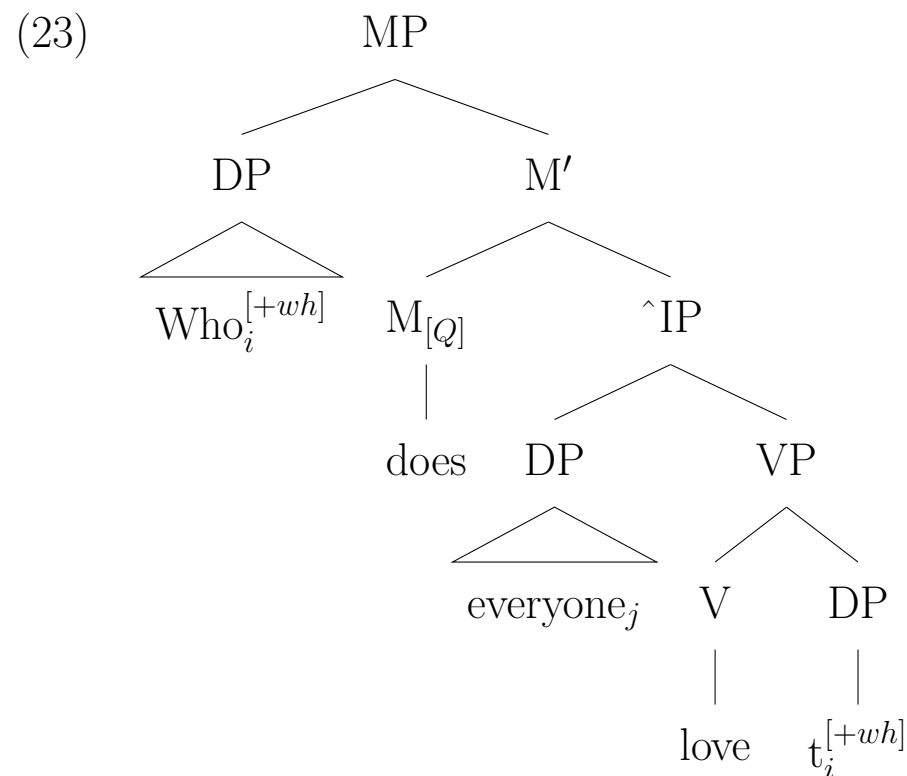
- (22) a. $\llbracket [_{DP} \textit{which}^{[+wh]} \alpha]_i^{[+wh]} \rrbracket^{\vec{g}g^*w^*} = \lambda \langle g, w \rangle \llbracket [_{DP} \textit{which}^{[+wh]} \alpha]_i \rrbracket^{\vec{g}gw}$
 $= \{ \lambda \langle g, w \rangle [\ll \pi \gg \vec{x}^{g[i/y]w}] : \vec{x} \in \textit{Alt}(\vec{g}(i)) \}$
 $\pi = \lambda \langle g', w' \rangle \exists y [\llbracket \alpha \rrbracket^{\vec{g}g'[i/y]w'} (\vec{x}^{g'[i/y]w'})]$, where $\langle g'[i/y], w' \rangle$ is in the domain of $\vec{x}$.
- b. π is accommodated in a linguistic context $g''w''$ in the following way:
 $\pi \Rightarrow \pi(\langle g'', w'' \rangle) = \exists y [\llbracket \alpha \rrbracket^{\vec{g}g''[i/y]w''} (\vec{x}^{g''[i/y]w''})]$ (contextualized)

Answer to Question 4: The scope of a *wh*-phrase depends on where the pre-supposition is accommodated.

3 Non-contextual meaning and functional *wh*-questions

3.1 Interpretation of a *wh*-question with the functional reading

- The structure of a *wh*-question with a functional reading:



- Compositional interpretation of a *wh*-question with the functional reading

$$\begin{aligned}
 (24) \quad & \text{a. } \llbracket t_i^{[+wh]} \rrbracket \vec{g}g^*w^* = \vec{g}(i)^{g^*w^*} \\
 & \text{b. } \llbracket t_j \text{ love } t_i^{[+wh]} \rrbracket \vec{g}g^*w^* = \text{love}^{w^*}(g^*(j), \vec{g}(i)^{g^*w^*}) \\
 & \text{c. } \llbracket \wedge [\text{everyone}_j [t_j \text{ love } t_i^{[+wh]}]] \rrbracket \vec{g}g^*w^* \\
 & \quad = \lambda w' [\lambda P \forall x [\text{person}^{w'}(x) \rightarrow P(x)] (\lambda z [\llbracket t_j \text{ love } t_i^{[+wh]} \rrbracket \vec{g}g^*[j/z]w'])] \\
 & \quad = \lambda w' \forall x [\text{person}^{w'}(x) \rightarrow \text{love}^{w'}(x, \vec{g}(i)^{g^*[j/x]w'})] \\
 & \text{d. } \llbracket \text{does } \wedge [\text{everyone}_j [t_j \text{ love } t_i^{[+wh]}]] \rrbracket \vec{g}g^*w^* \\
 & \quad = \langle \lambda w' \forall x [\text{person}^{w'}(x) \rightarrow \text{love}^{w'}(x, \vec{g}(i)^{g^*[j/x]w'})], \\
 & \quad \quad \lambda w' \neg \forall x [\text{person}^{w'}(x) \rightarrow \text{love}^{w'}(x, \vec{g}(i)^{g^*[j/x]w'})] \rangle \\
 & \text{e. } \llbracket \text{Who}_i^{[+wh]} \rrbracket \vec{g}g^*w^* = \llbracket [\text{Which}^{[+wh]} \text{ person}]_i^{[+wh]} \rrbracket \vec{g}g^*w^* \\
 & \quad = \{ \lambda \langle g, w \rangle [\ll \pi \gg \vec{x}^{g[i/y]w}] : \vec{x} \in \text{Alt}(\vec{g}(i)) \}, \text{ where} \\
 & \quad \quad \pi: \lambda \langle g'', w'' \rangle \exists y [\text{person}^{w''}(\vec{x}^{g''[i/y]w''})] \\
 & \text{f. } \llbracket \text{Who}_i^{[+wh]} \text{ does } \wedge [\text{everyone}_j [t_j \text{ love } t_i^{[+wh]}]] \rrbracket \vec{g}g^*w^* = \\
 & \quad \{ \lambda \vec{z} [\llbracket \text{does } \wedge [\text{everyone}_j [t_j \text{ love } t_i^{[+wh]}]] \rrbracket \vec{g}[i/\vec{z}]g^*w^*(\vec{x}) \mid \vec{x} \in \llbracket \text{Who}_i^{[+wh]} \rrbracket \vec{g}g^*w^* \} \\
 & \quad = \text{POW}(\{ \lambda w' \forall x [\text{person}^{w'}(x) \rightarrow \text{love}^{w'}(x, \ll \pi \gg \vec{x}^{g^*[i/y][j/x]w'})] \mid \vec{x} \in \text{Alt}(\vec{g}(i)) \})
 \end{aligned}$$

Functional reading

$$\begin{aligned}
 (25) \quad & POW(\{\lambda w' \forall x [person^{w'}(x) \rightarrow \ll \pi(\langle g^*[j/x], w' \rangle) \gg \\
 & \quad \quad \quad love^{w'}(x, \vec{x}^{g^*[i/y][j/x]w'}) \mid \vec{x} \in Alt(\vec{g}(i)) \}) \\
 & = POW(\{\lambda w' \forall x [person^{w'}(x) \rightarrow \ll \exists y [person^{w'}(\vec{x}^{g^*[i/y][j/x]w'})] \gg \\
 & \quad \quad \quad love^{w'}(x, \vec{x}^{g^*[i/y][j/x]w'}) \mid \vec{x} \in Alt(\vec{g}(i)) \}), \text{ where} \\
 & \ll \pi(\langle g^*[j/x], w' \rangle) \gg = \lambda \langle g'', w'' \rangle \exists y [person^{w''}(\vec{x}^{g''[i/y]w''})] (\langle g^*[j/x], w' \rangle) \\
 & = \exists y [person^{w'}(\vec{x}^{g^*[i/y][j/x]w'})]
 \end{aligned}$$

- The speaker assumes a set of alternatives $\vec{x}$ that look like the one in (26).

$$(26) \quad \vec{x} = \left[\begin{array}{ccc} g_1 w' & \rightarrow & c_1 \\ g_2 w' & \rightarrow & c_2 \\ g_3 w' & \rightarrow & c_3 \\ \dots & \dots & \dots \end{array} \right], \text{ where } g_1, g_2, g_3, \dots \text{ are s.t. } \begin{array}{cccccc} & & & & [i/y] & \dots & [j/x] & \dots \\ \hline g_1 & \rightarrow & c_1 & \dots & a_1 & \dots \\ g_2 & \rightarrow & c_2 & \dots & a_2 & \dots \\ g_3 & \rightarrow & c_3 & \dots & a_3 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

(c_n is in a relation of natural function to a_n .)

The universal quantifier and then the *wh*-phrase modify the assignments so that the resulting assignments are in the domain of $\vec{x}$

Individual reading

$$\begin{aligned}
 (27) \quad & POW(\{\lambda w' \ll \lambda \langle g'', w'' \rangle \exists y [person^{w''}(\vec{x}^{g''[i/y]w''})](\langle g^*, w' \rangle) \gg \forall x [person^{w'}(x) \rightarrow \\
 & \quad love^{w'}(x, \vec{x}^{g^*[i/y][j/x]w'})] \mid \vec{x} \in Alt(\vec{g}(i))\}) \\
 & = POW(\{\lambda w' \ll \exists y [person^{w'}(\vec{x}^{g^*[i/y]w'})] \gg \forall x [person^{w'}(x) \rightarrow \\
 & \quad love^{w'}(x, \vec{x}^{g^*[i/y][j/x]w'})] \mid \vec{x} \in Alt(\vec{g}(i))\})
 \end{aligned}$$

- The speaker assumes a set of alternatives $\vec{x}$ that look like the one in (28).

$$(28) \quad \vec{x} = \begin{bmatrix} g_1 w' \rightarrow c_1 \\ g_2 w' \rightarrow c_1 \\ g_3 w' \rightarrow c_1 \\ \dots \quad \dots \quad \dots \end{bmatrix}, \text{ where } g_1, g_2, g_3, \dots \text{ are s.t. } \begin{array}{cccccc} & & & [i/y] & \dots & [j/x] & \dots \\ & & & \hline g_1 & \rightarrow & c_1 & \dots & a_1 & \dots \\ g_2 & \rightarrow & c_1 & \dots & a_2 & \dots \\ g_3 & \rightarrow & c_1 & \dots & a_3 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

The *wh*-phrase modifies the assignments first so that it gets a fixed value, and then the universal quantifier modifies the modified assignments again. The resulting assignments are in the domain of $\vec{x}$.

3.2 Problems with a multiply-indexed trace solved

- The meaning of personhood of *who* is in the meaning of *person*^{w'}($\vec{x}^{g^*[i/y][j/x]w'}$) in (24).

$$(24) \quad f. \quad \llbracket \text{Who}_i^{[+wh]} \text{ does } \wedge [\text{everyone}_j [t_j \text{ love } t_i^{[+wh]}]] \rrbracket \vec{g}g^*w^* = \\ POW(\{ \lambda w' \forall x [person^{w'}(x) \rightarrow \ll \exists y [person^{w'}(\vec{x}^{g^*[i/y][j/x]w'})] \gg \\ love^{w'}(x, \vec{x}^{g^*[i/y][j/x]w'})] \mid \vec{x} \in Alt(\vec{g}(i)) \})$$

- A non-contextual value of a trace with one index includes in its domain all possible contextual assignments which assign values to indexes of any number of DPs.

- (3) a. Q: To whom does every writer dedicate a book?
A: To the person that inspired her/him to write it.
- b. $\llbracket [\text{to whom}]_i [\text{every writer}]_j [\text{a book}]_k [t_j \text{ dedicates } t_k t_i^{j,k}] \rrbracket$
- c. $\{p \mid \exists f[p = \forall x[writer(x) \rightarrow \exists y[book(y) \wedge dedicate(x, y, f(x, y))]]]\}$

$$(29) \quad POW(\{\lambda w' \forall x [writer^{w'}(x) \rightarrow \exists y [book^{w'}(y) \wedge \ll \exists z [person^{w'}(\vec{x}^{g^*[i/z][j/x][k/y]w'})] \gg \\ dedicate^{w'}(x, y, \vec{x}^{g^*[i/z][j/x][k/y]w'})]] | \vec{x} \in Alt(\vec{g}(i))\})$$

$$\vec{x} = \begin{bmatrix} g_1 w' \rightarrow c_1 \\ g_2 w' \rightarrow c_2 \\ g_3 w' \rightarrow c_3 \\ \dots \dots \dots \end{bmatrix}, \text{ where } g_1, g_2, g_3, \dots \text{ are s.t.}$$

		$[i/z]$	$\dots$	$[j/x]$	$\dots$	$[k/y]$	$\dots$
g_1	$\rightarrow$	c_1	$\dots$	a_1	$\dots$	b_1	$\dots$
g_2	$\rightarrow$	c_2	$\dots$	a_2	$\dots$	b_2	$\dots$
g_3	$\rightarrow$	c_3	$\dots$	a_3	$\dots$	b_3	$\dots$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

$g_n(i)$ is in a functional relation to $\langle g_n(j), g_n(k) \rangle$.

- A short answer can be interpreted with no free variable.

$$(30) \quad \text{The meaning of the short answer:}$$

$\llbracket \text{Their}_j \text{ mother} \rrbracket^{\vec{g}g^*w^*} = \lambda \langle g', w' \rangle \iota z [mother^{w'}(z, g'(j))]$, where there is no free variable.

4 Implications

- Non-contextual meanings are necessary in the semantics for natural language, together with stages, objects, kinds, individual concepts, and characters.
- Syntactic binding should be replaced with semantic binding.
- If an expression generates alternatives, like a focused phrase, it denotes a non-contextual meaning.
- No attempt to explain the exceptional scope of an indefinite based on alternative semantics is in the right direction because alternatives are scopeless.